Replication: Monitoring Corruptible Politicians (Bobonis et al. 2016)

ECON 544: Political Economy

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1 Introduction

1.1 Context

This paper replicates and extends the empirical analysis of Bobonis et al. (2016), which studies how information about corruption affects electoral accountability in Puerto Rico. The original paper exploits variation in the timing of municipal audits to identify the causal impact of corruption disclosure on both electoral outcomes and subsequent corruption behavior.

A key feature of the setting is that audit timing is plausibly exogenous to current political conditions. Although municipalities are audited in a predetermined order, delays in the audit process generate quasi-random variation in whether audit results are released before or after municipal elections. This allows the authors to compare municipalities where corruption information becomes available to voters before elections ("timely audits") to those where it is released afterward ("untimely audits").

The main findings of the original paper are twofold. First, while timely audits reduce corruption in the period leading up to elections, they do not lead to sustained reductions in corruption in subsequent terms. Instead, the evidence suggests that voters may reelect "competent but corruptible" politicians (i.e., incumbents who respond to electoral incentives but do not remain less corrupt in the long run). Second, the release of audit information prior to elections reduces incumbent reelection probabilities, particularly when audits reveal higher levels of corruption.

This replication reproduces the core results of the paper using the original data and empirical specifications. In addition, the paper introduces two extensions:

- The first examines whether the effects of audit disclosure vary with voter education levels, testing for heterogeneity in electoral accountability. The central hypothesis is that more educated voters may be better able to process and act upon information revealed by audits (Glaeser et al., 2007; Botero et al., 2013), leading to stronger electoral accountability effects (Besley, 2006; Ferraz & Finan, 2008).

- The second applies a modern difference-in-differences estimator designed for settings with non-absorbing treatments, reassessing the dynamic effects of audit timing on corruption outcomes. Recent econometric research has shown that traditional two-way fixed effects (TWFE) estimators may produce biased estimates when treatment timing varies across units and treatment effects evolve over time (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfoeuille, 2020). Given that municipalities in the audit program receive timely audits in different election cycles, and that Bobonis et al. (2016)'s theoretical model predicts dynamic effects, this extension provides an important robustness check of the original findings.

1.2 Data

Bobonis et al. (2016) construct a longitudinal dataset of municipal government audits in Puerto Rico from 1987 to 2005. The Office of the Comptroller of Puerto Rico (OCPR) audits municipalities in a pre-established and fixed order established in the 1950s, making the timing of audits plausibly exogenous (Bobonis et al., 2016) . Following the original authors, I define a "timely audit" as one whose report is published in the two-year period preceding a mayoral election. Audits published in the two-year period following an election are classified as "untimely."

The main outcome variables are measures of corruption based on audit findings: the number of corrupt violations per report (total, attributed to the mayor/vice-mayor, and referred to the Department of Justice), and the share of findings classified as corrupt. Electoral outcomes include whether the incumbent runs for reelection, wins reelection, and wins conditional on running. Data on municipality characteristics come from the 1990 US Census.

For my replication and extensions, I use only the data within the replication file for this study (Bobonis et. al, 2019).

1.3 Methodology

Bobonis et al. (2016) estimate the effect of timely audit disclosure on corruption outcomes using the following two-way fixed effects specification:

$$c_{m,t} = \theta \mathrm{A}_{m,t} + \beta X_{m,t} + \alpha_m + \gamma_t + \varepsilon_{m,t} \tag{1}$$

where $c_{m,t}$ is a corruption measure in municipality m around election year t, $A_{m,t}$ is an indicator for whether the audit report was published in the two-year period preceding election t, $X_{m,t}$ is a vector of controls (number of reports, party indicators, opposition to governor or comptroller dummy, previous election vote share, terms in office), and γ_t and α_m are election year and municipality fixed effects. Standard errors are clustered at the municipality level.

Furthermore, their second main specification estimates the effect of a timely audit on electoral outcomes:

$$e_{m,t} = \theta_{E1} A_{m,t} + \theta_{E2} A_{m,t} c_{m,t} + \beta_{E1} c_{m,t} + \beta_{E2} X_{m,t} + \gamma_t + \alpha_m + \epsilon_{m,t} \tag{2}$$

where $e_{m,t}$ indicates whether the incumbent runs for reelection, wins reelection, or wins conditional on running. These estimates capture equilibrium relationships rather than causal effects.

To examine the longer-term consequences of audits on rent-seeking behavior, the authors estimate the effect of the period t audits on the reported corruption levels in the subsequent audit (at t+1):

$$c_{m,t+1} = \theta_{P1} A_{m,t} + \theta_{P2} A_{m,t+1} + \beta X'_{m,t} + \gamma_{t+1} + \alpha_m + \epsilon_{m,t+1} \tag{3}$$

where $c_{m,t+1}$ denotes the corruption outcome in the subsequent audit, and $A_{m,t+1}$ is a control for the next audit.

Furthermore, to examine the consequences of a timely audit on the next-period incumbent's electoral outcomes, the authors estimate:

$$e_{m,t+1} = \theta_{E'1} A_{m,t} + \beta_{E'1} A_{m,t+1} + \beta_{E'2} X_{m,t} + \gamma_{t+1} + \alpha_m + \epsilon_{m,t+1} \tag{4}$$

2 Replication

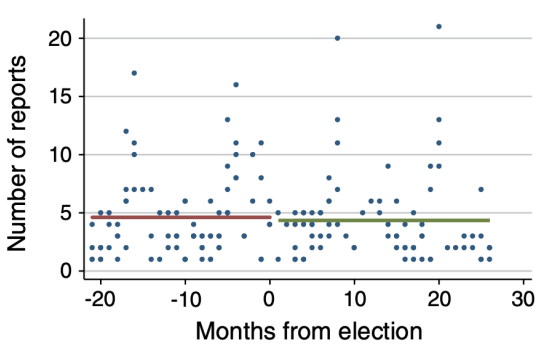
In this section, I replicate the results of the original study and present them next to each other. For the rest of this section, columns labeled "A" are the authors' original results and columns "B" are my replication results. Differences between the two are bold.

2.1 Background

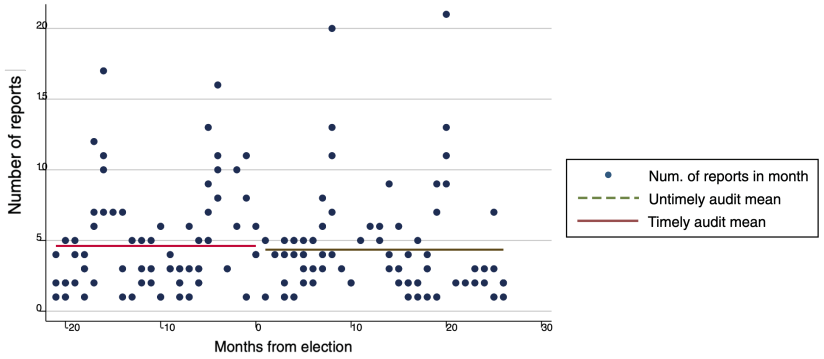
2.1.1 The OCPR Municipal Government Auditing Program

In Figure 1, the timing of the dissemination of audits is depicted. Due to the OCPR's resource constraints, there were delays in the publication of audits, despite their prespecified order. Figure 1 shows there is no evidence of bias in the timing of the publication of the reports before an election (panel A) nor by whether the incumbent mayor is in opposition to the governor in office or the party that appointed the comptroller (panel B). There are no significant differences in the authors' graphs and those I replicated.

Panel A. Number of corrupt violations, per report

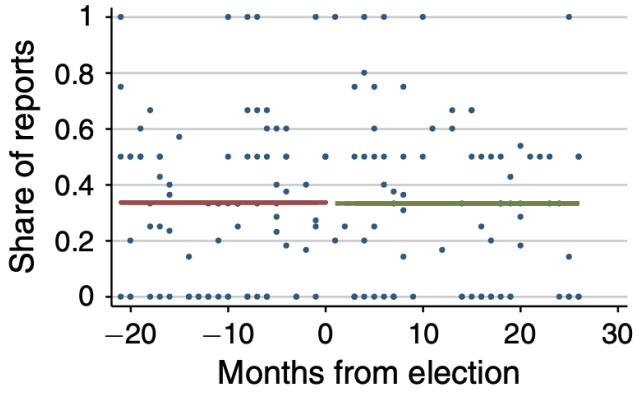


(a) Original

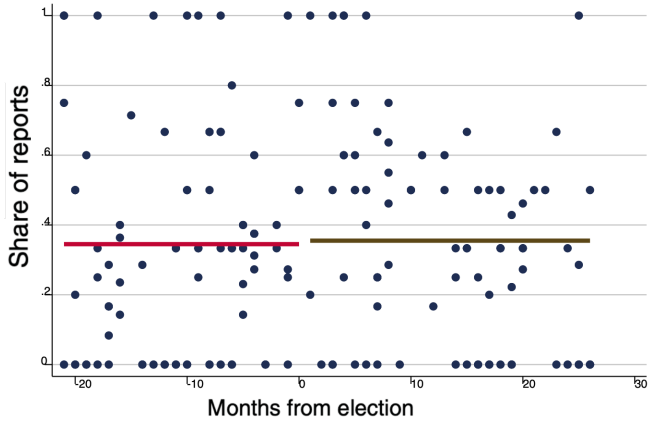


(b) Replication

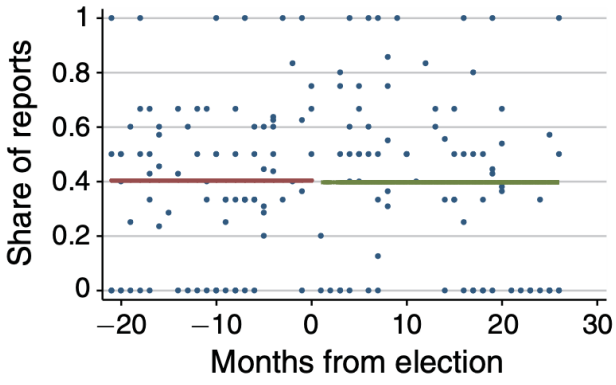
Panel B. Share of reports on municipalities where incumbent is from the opposition party to governor / comptroller



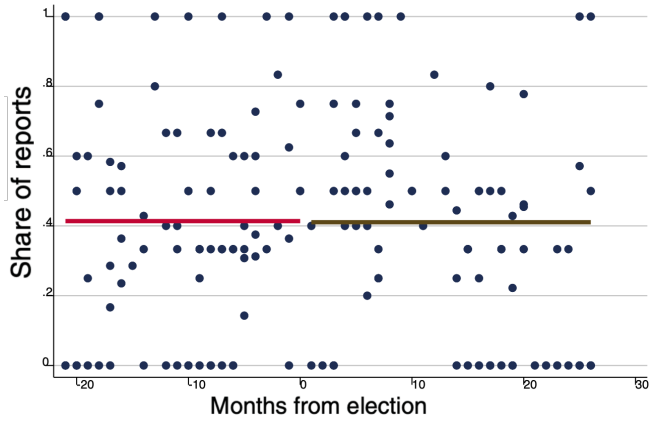
(c) Original



(d) Replication



(e) Original



(f) Replication

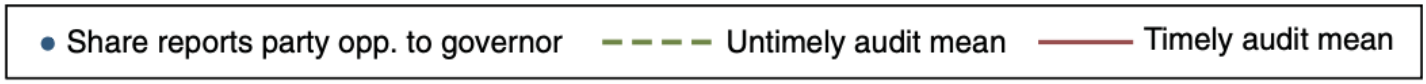


Figure 1: Publication of Audit Reports, Timing and Outcomes, 1987–2005

Notes: Panel A shows the timing of release of the number of reports by month in the four-year period around each election (in November 1988, 1992, 1996, 2000, and 2004). Panel B presents the share of published reports of municipalities in which the incumbent is in the opposition party to the governor in office or to the governor who appointed the comptroller in office, in each month. The red line in each figure demarcates the mean for the 22 months before the November election; the green lines demarcate the mean for the 26 months following an election. The original results from Bobonis et al. (2016) (a, c, e) are presented next to my replication results (b, d, f) and show no differences. Any differences may be attributed to colour schemes and scaling of the images.

2.2 Data

2.2.1 Measures of Corruption Based on the Audit Reports

Table 1 presents the means and standard deviations of audit outcome variables (panel A) and characteristics of the audits (panel B). There is a small (≈ 0.01) difference in a standard error for the "Any corrupt violation" variable (column 4); this may be attributed to a rounding error in the construction of the original table since none of the other replication results differ from the original.

Table 1: Characteristics of the Audit Report

	All (1)		Timely audit (3)		Untimely audit (2)		Difference (4)		Observations (5)	
Variable	A	B	A	B	A	B	A	B	A	B
Panel A. Audit outcomes										
Any corrupt violation [1/0]	0.69	0.69	0.62	0.62	0.78	0.78	-0.13	**-0.12**	326	326
	[0.46]	[0.46]	[0.49]	[0.49]	[0.41]	[0.41]	(0.05)	**(0.058)**		
Number of all corrupt violations per report	1.38	1.38	0.80	0.80	2.17	2.17	-1.36	-1.36	326	326
	[1.79]	[1.79]	[0.99]	[0.99]	[2.26]	[2.26]	(0.21)	(0.21)		
Number of violations by mayor/vice-mayor	0.58	0.58	0.31	0.31	0.95	0.95	-0.60	-0.60	326	326
	[0.99]	[0.99]	[0.57]	[0.57]	[1.28]	[1.28]	(0.12)	(0.12)		
Number of violations referred to DOJ	0.65	0.65	0.40	0.40	0.99	0.99	-0.67	-0.67	326	326
	[1.20]	[1.20]	[0.81]	[0.81]	[1.52]	[1.52]	(0.16)	(0.16)		
Share of findings, attributed to mayor/ vice-mayor, classified as corrupt	0.19	0.19	0.16	0.16	0.22	0.22	-0.06	-0.06	326	326
	[0.28]	[0.28]	[0.29]	[0.29]	[0.26]	[0.26]	(0.04)	(0.04)		
Share of findings referred to DOJ classified as corrupt	0.27	0.27	0.22	0.22	0.34	0.34	-0.11	-0.11	326	326
	[0.37]	[0.37]	[0.35]	[0.35]	[0.39]	[0.39]	(0.05)	(0.05)		
Panel B. Other audit characteristics										
Number of audit reports	1.88	1.88	2.16	2.16	1.49	1.49	0.77	0.77	326	326
	[1.15]	[1.15]	[1.32]	[1.32]	[0.72]	[0.72]	(0.13)	(0.13)		
Start of audit period in reports (years from election)	6.31	6.31	6.84	6.84	5.59	5.59	1.22	1.22	326	326
	[2.59]	[2.59]	[2.70]	[2.70]	[2.25]	[2.25]	(0.25)	(0.25)		
End of audit period in reports (years from election)	1.24	1.24	1.56	1.56	0.80	0.80	0.43	0.43	326	326
	[1.60]	[1.60]	[1.57]	[1.57]	[1.54]	[1.54]	(0.16)	(0.16)		
Time span of audited period [years]	5.08	5.08	5.29	5.29	4.79	4.79	0.78	0.78	326	326
	[2.59]	[2.59]	[2.76]	[2.76]	[2.32]	[2.32]	(0.24)	(0.24)		
Deviation from predetermined order of reports										
Mean of deviation from expected number of reports	-0.16	-0.16	1.04	1.04	-1.48	-1.48	3.38	3.38	220	220
	[16.91]	[16.91]	[18.24]	[18.24]	[15.28]	[15.28]	(2.64)	(2.64)		
Mean of absolute value of deviation from expected number of reports	14.30	14.30	15.79	15.79	12.66	12.66	0.70	0.70	220	220
	[10.88]	[10.88]	[11.03]	[11.03]	[10.52]	[10.52]	(1.88)	(1.88)		

Notes: Standard deviations of variables are reported in brackets. Differences estimated in ordinary least squares (OLS) regression models, regression-adjusted for municipality and electoral term fixed effects. Robust standard errors of mean differences are reported in parentheses. Columns labeled **A** represent the original results (Bobonis et al., 2016); columns labeled **B** represent the replication results. Bolded results indicate differences between the original and replication results.

2.2.2 Other Data Sources

Table 2 presents descriptive statistics for the outcome and control variables that the authors use in their specification. The results show there are no significant differences between municipalities that receive timely and untimely audits across many mayoral, political, and socioeconomic characteristics.

There is one small difference between the original and replication results. The standard deviation of the opposing party dummy variable (column 3) is different from the original results (≈ 0.01). This difference is small which suggests a possible rounding error in the original construction of the table. Moreover, these errors are not found in the rest of the table.

Table 2: Characteristics of the Municipalities

	All (1)		Timely audit (2)		Untimely audit (3)		Difference (adjusted) (4)		Observations (5)	
	A	B	A	B	A	B	A	B	A	B
Panel A. Electoral outcomes										
Incumbent runs for reelection (1/0)	0.74	0.74	0.72	0.72	0.76	0.76	−0.03	−0.03	326	326
	[0.44]	[0.44]	[0.45]	[0.45]	[0.43]	[0.43]	(0.06)	(0.06)		
Incumbent party wins (1/0)	0.33	0.33	0.35	0.35	0.31	0.31	−0.05	−0.05	326	326
	[0.47]	[0.47]	[0.48]	[0.48]	[0.46]	[0.46]	(0.05)	(0.05)		
Incumbent mayor wins | running (1/0)	0.35	0.35	0.37	0.37	0.32	0.32	−0.11	−0.11	241	241
	[0.48]	[0.48]	[0.48]	[0.48]	[0.47]	[0.47]	(0.06)	(0.06)		
Panel B. Incumbent mayor characteristics										
Mayor, member of PNP (1/0)	0.51	0.51	0.52	0.52	0.49	0.49	0.09	0.09	326	326
	[0.50]	[0.50]	[0.50]	[0.50]	[0.50]	[0.50]	(0.06)	(0.06)		
Member of opposition party to governor (1/0)	0.33	0.33	0.35	0.35	0.30	0.30	0.04	0.04	326	326
	[0.47]	[0.47]	[0.48]	[0.48]	[0.46]	[0.46]	(0.06)	(0.06)		
Member of opp. party to governor appointing comptroller (1/0)	0.44	0.44	0.47	0.47	0.41	0.41	0.06	0.06	326	326
	[0.50]	[0.50]	[0.50]	[0.50]	[0.50]	**[0.49]**	(0.07)	(0.07)		
Terms in office	1.15	1.15	1.19	1.19	1.09	1.09	0.07	0.07	326	326
	[1.24]	[1.24]	[1.36]	[1.36]	[1.07]	[1.07]	(0.14)	(0.14)		
Mayor's win margin in previous election	0.11	0.11	0.11	0.11	0.10	0.10	0.00	0.00	326	326
	[0.09]	[0.09]	[0.09]	[0.09]	[0.09]	[0.09]	(0.01)	(0.01)		
Party incumbency advantage (1/0)	0.55	0.55	0.57	0.57	0.52	0.52	0.04	0.04	326	326
	[0.50]	[0.50]	[0.50]	[0.50]	[0.50]	[0.50]	(0.06)	(0.06)		
Panel C. Pre-audit municipality characteristics										
Share of pop. high school education or more (1990)	0.44	0.44	0.44	0.44	0.44	0.44	0.01	0.01	326	326
	[0.08]	[0.08]	[0.07]	[0.07]	[0.08]	[0.08]	(0.01)	(0.01)		
Share of pop. college education or more (1990)	0.10	0.10	0.11	0.11	0.10	0.10	0.00	0.00	326	326
	[0.04]	[0.04]	[0.04]	[0.04]	[0.04]	[0.04]	(0.01)	(0.01)		
Household median income (US$1,000s) (1990)	8.22	8.22	8.26	8.26	8.16	8.16	0.12	0.12	326	326
	[1.75]	[1.75]	[1.75]	[1.75]	[1.77]	[1.77]	(0.24)	(0.24)		
Poverty rate (1990)	0.61	0.61	0.61	0.61	0.62	0.62	−0.01	−0.01	326	326
	[0.10]	[0.10]	[0.10]	[0.10]	[0.09]	[0.09]	(0.01)	(0.01)		
Unemployment rate (first year of mayor's term)	0.17	0.17	0.16	0.16	0.18	0.18	0.00	0.00	216	216
	[0.05]	[0.05]	[0.05]	[0.05]	[0.06]	[0.06]	(0.00)	(0.00)		
Unemployment rate (second year of mayor's term)	0.16	0.16	0.16	0.16	0.17	0.17	0.00	0.00	273	273
	[0.05]	[0.05]	[0.05]	[0.05]	[0.05]	[0.05]	(0.00)	(0.00)		
Panel D. Municipal government budget										
Total expenditures (first and second year of mayor's term)	11.76	11.76	13.62	13.62	8.27	8.27	−0.21	−0.21	216	216
	[23.32]	[23.32]	[27.60]	[27.60]	[10.96]	[10.96]	(0.28)	(0.28)		
Capital improvements	0.22	0.22	0.22	0.22	0.22	0.22	0.00	0.00	212	212
	[0.43]	[0.43]	[0.42]	[0.42]	[0.45]	[0.45]	(0.02)	(0.02)		
Salaries and benefits	4.82	4.82	4.99	4.99	4.51	4.51	−0.07	−0.07	212	212
	[5.38]	[5.38]	[5.32]	[5.32]	[5.53]	[5.53]	(0.24)	(0.24)		
Social assistance	0.23	0.23	0.22	0.22	0.25	0.25	0.04	0.04	212	212
	[0.38]	[0.38]	[0.36]	[0.36]	[0.41]	[0.41]	(0.04)	(0.04)		
Other expenditures	4.01	4.01	4.39	4.39	3.33	3.33	0.14	0.14	212	212
	[5.60]	[5.60]	[5.90]	[5.90]	[4.97]	[4.97]	(0.20)	(0.20)		
Property tax	3.25	3.25	3.49	3.49	2.82	2.82	0.26	0.26	212	212
	[5.50]	[5.50]	[5.87]	[5.87]	[4.77]	[4.77]	(0.21)	(0.21)		
Licensing	1.95	1.95	2.08	2.08	1.71	1.71	−0.13	−0.13	212	212
	[3.31]	[3.31]	[3.23]	[3.23]	[3.47]	[3.47]	(0.09)	(0.09)		
Waste disposal	0.22	0.22	0.27	0.27	0.12	0.12	−0.01	−0.01	212	212
	[0.79]	[0.79]	[0.95]	[0.95]	[0.33]	[0.33]	(0.03)	(0.03)		
Transfers and other revenue	3.01	3.01	3.02	3.02	2.99	2.99	−0.21	−0.21	216	216
	[2.15]	[2.15]	[1.97]	[1.97]	[2.48]	[2.48]	(0.33)	(0.33)		

Notes: Standard deviations of variables are reported in brackets. Differences estimated in ordinary least squares (OLS) regression models, regression-adjusted for municipality and electoral term fixed effects (a = except pre-audit municipality characteristics differences (data available from 1990 decennial census)). Robust standard errors of mean differences are reported in parentheses. Columns labeled **A** represent the original results (Bobonis et al., 2016); columns labeled **B** represent the replication results. Bolded results indicate differences between the original and replication results.

2.3 Methodology

Figure 2 is a simple illustration of the timing of the audits around election periods, where timely audits are those released two years before election t, and untimely ones are released in the two years after. This is the authors' original figure; there was no replication code to reproduce it.

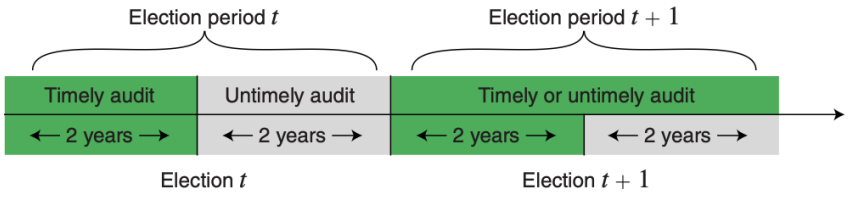


Figure 2: Empirical Framework

Notes: A municipality with a timely audit during election period t may have a timely or untimely audit during election period $t+1$. In this sample, 73% of municipalities with a timely audit at time t received a timely audit the following election period $(t+1)$, whereas 54% of those with an untimely audit have a timely audit during election period $(t+1)$. Figure directly extracted from Bobonis et al. (2016) since no replication code was provided by the authors.

Figure 3 plots the number of findings (corrupt in Panel A, non-corrupt in Panel B) each year relative to the first year of the mayor's term. In Panel A, there is no significant difference in corrupt findings during the second year of the term. However, the number of corrupt violations drops significantly during the last two years of the term. Panel B shows no similar pattern for non-corrupt findings. My replication results are visually identical to the originals.

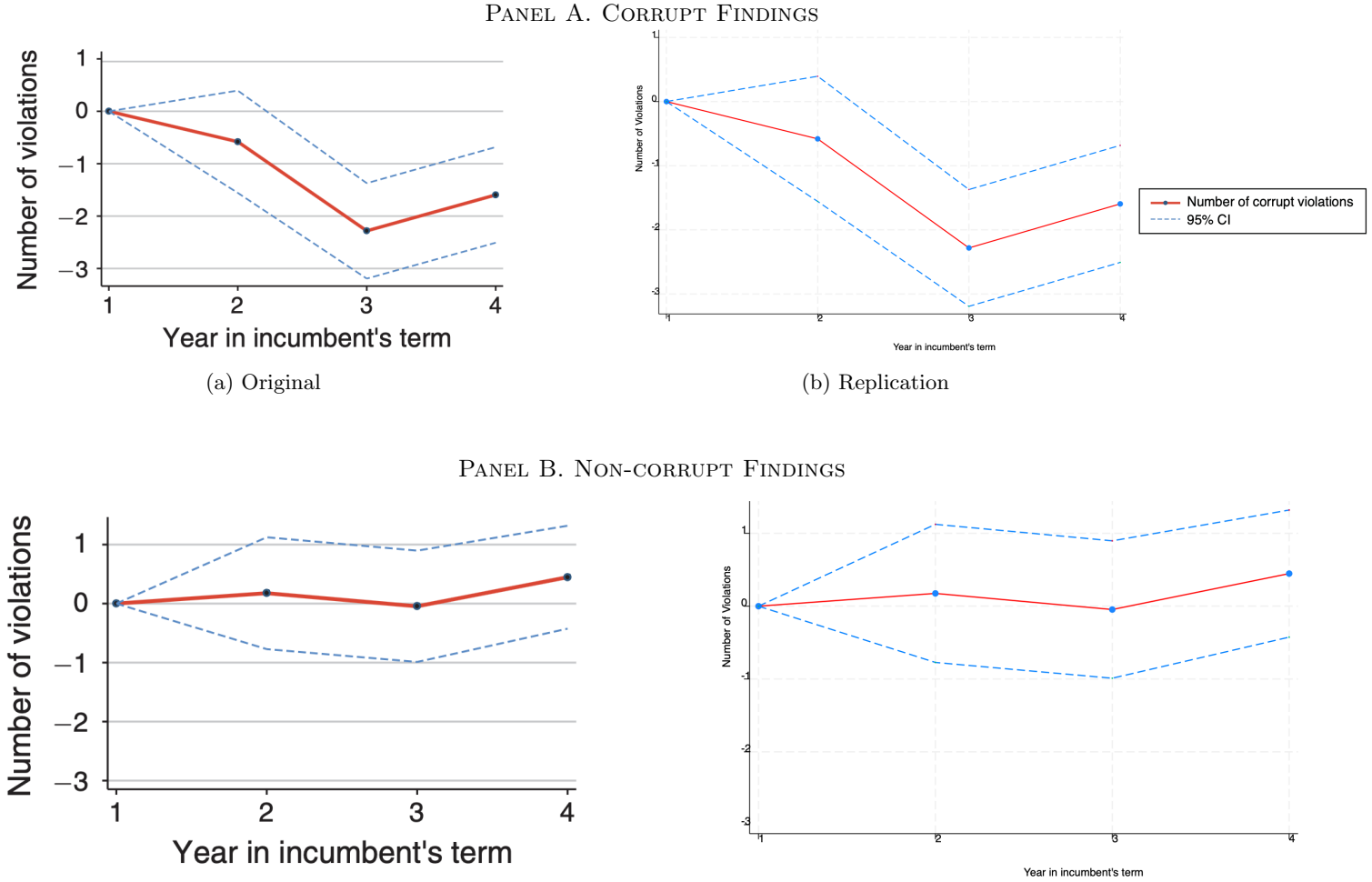


Figure 3: Number of Findings in Reports by Year of Publication during Incumbent's Term

Notes: Panel A shows the differences in the number of findings classified as corrupt in audit reports by the year of publication within an incumbent mayor's term, relative to the first year of the term, together with 95% confidence intervals (regression-adjusted for election cycle fixed effects). Panel B shows the analogous relationship for findings classified as not corrupt. The original results from Bobonis et al. (2016) (a, c) are presented next to my replication results (b, d) and show no differences.

2.4 Results

2.4.1 Effects of the Audits on Corruption

Figure 4 is a plot of corruption outcomes found in each election term, relative to election term t. Trends in corruption are shown for timely and untimely audits. This relationship is shown for all corruption violations per report (panel A), the subset attributed to the mayor/vice-mayor (panel B), and the number referred to the Department of Justice (DOJ) (panel C). The authors find there are no differences across outcomes in earlier audits, but there is a drop in corruption around election t. This effect rebounds in later audits (election year $t+4$). There are no differences between the original and replication results in these figures.

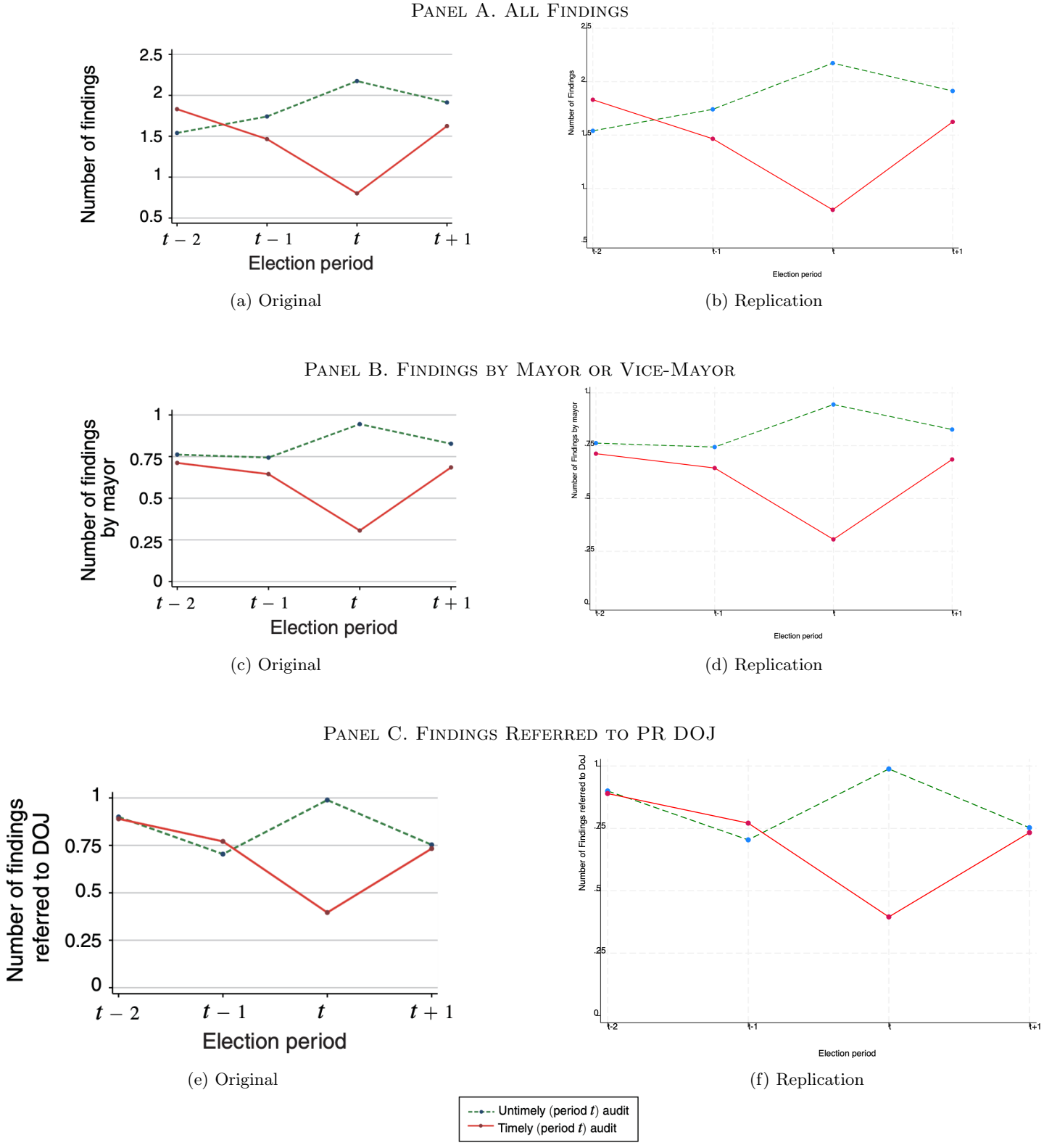


Figure 4: Number of Findings Across Time, By Timeliness of Audit in Election (t)

Notes: The original results from Bobonis et al. (2016) (a, c, e) are presented next to my replication results (b, d, f) and show no differences.

Table 3 presents the baseline short-term deterrence effects of the monitoring program. The authors estimate that the timely dissemination of an audit prior to an election (t) leads to a statistically significant decrease of 1.43 corrupt violations per report, which constitutes an approximate 67 percent reduction in rent-seeking behavior compared to municipalities that received untimely audits. This finding is highly robust across alternative measures of corruption, including a 0.65 reduction in severe violations referred to the Department of Justice (DOJ). Table 3, panel B shows that the disciplining effect of audits is higher in municipalities with competitive elections. The replication successfully reproduces these point estimates side-by-side without any discrepancies, confirming the presence of a strong short-term sanctioning effect where incumbent mayors temporarily restrain their corrupt activities in anticipation of electoral sanctioning.

Table 3: Effects of the (Timing of the) Audits on the Number of Corrupt Violations in the Current Audit

	Number of corrupt violations, per report						Share of findings classified as corrupt			
	All (1)		By mayor/vice-mayor (2)		Ref. to DOJ (3)		By mayor/vice-mayor (4)		Ref. to DOJ (5)	
	A	B	A	B	A	B	A	B	A	B
Panel A. Average effects										
Timely audit	-1.43	-1.43	-0.63	-0.63	-0.65	-0.65	-0.082	-0.082	-0.160	-0.160
	(0.22)	(0.22)	(0.14)	(0.14)	(0.16)	(0.16)	(0.040)	(0.040)	(0.055)	(0.055)
Mun. controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year & Mun. FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Panel B. Effects by Party Advantage										
Timely audit	-1.90	-1.90	-0.92	-0.92	-1.12	-1.12	-0.174	-0.174	-0.229	-0.229
	(0.32)	(0.32)	(0.17)	(0.17)	(0.25)	(0.25)	(0.045)	(0.045)	(0.069)	(0.069)
Timely audit × party advantage	0.96	0.96	0.61	0.61	0.97	0.97	0.191	0.191	0.153	0.153
	(0.41)	(0.41)	(0.24)	(0.24)	(0.32)	(0.32)	(0.061)	(0.061)	(0.106)	(0.106)
Mun. controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year & Mun. FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	326	326	326	326	326	326	326	326	326	326
Mean of dep. variable (untimely audits)	2.17	2.17	0.95	0.95	0.99	0.99	0.22	0.22	0.34	0.34

Notes: Coefficient estimates and standard errors from OLS regressions are presented; disturbance terms are clustered at the municipality level. Controls are the number of municipality government reports, the number of municipal public corporation or consortium reports; indicators for New Progressive Party membership, for incumbent in the opposition party to the state-level executive government, and for incumbent in the opposition party to the governor who appointed comptroller; the vote share for the incumbent in the previous election ($t-1$); and the incumbent's number of terms in office (at time t). The sample is composed of all municipalities in which mayors are running for reelection that had a first audit during 1987 to 2002. Columns labeled **A** represent the original results (Bobonis et al., 2016); columns **B** represent the replication results. There are no differences between the results.

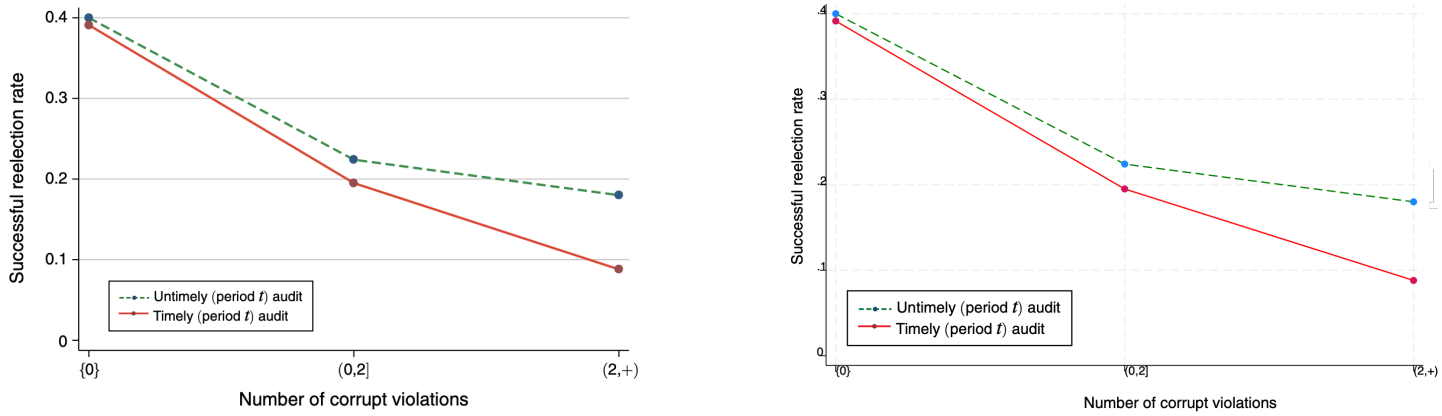
Table 4 tests whether there is a sustained decrease in corruption in the subsequent term $(t+1)$. The estimates reveal that there is no statistically significant reduction in long-term rent-seeking. For the most stringent measure (findings referred to the DOJ), the point estimate implies a (statistically insignificant) increase of 0.16 violations per report (column 3). The replication results are exactly the same.

Table 4: The Effects of the Audits on the Number of Corrupt Violations in the Subsequent Audit *(Term)*

	Number of corrupt violations, per report in next audit						Share of findings classified as corrupt in next audit			
	All (1)		By mayor/vice-mayor (2)		Ref. to DOJ (3)		By mayor/vice-mayor (4)		Ref. to DOJ (5)	
	A	B	A	B	A	B	A	B	A	B
Panel A. All Municipalities										
Timely audit (at t)	-0.10	-0.10	-0.07	-0.072	0.16	0.16	0.058	0.058	0.032	0.032
	(0.32)	(0.32)	(0.16)	(0.16)	(0.24)	(0.24)	(0.055)	(0.055)	(0.067)	(0.067)
Municipality controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Election Year & Mun. FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	232	232	232	232	232	232	232	232	232	232
Mean of dep. variable (post-election)	1.91	1.91	0.83	0.83	0.75	0.75	0.21	0.21	0.36	0.36

Notes: Coefficient estimates and standard errors from OLS regressions are presented; disturbance terms are clustered at the municipality level. Controls are the number of municipality government reports, the number of municipal public corporation or consortium reports; indicators for New Progress Party membership, for incumbent in the opposition party to the state-level executive government, and for incumbent in the opposition party to the governor who appointed comptroller; the vote share for the incumbent in the previous election ($t-1$); and the incumbent's number of terms in office (at time t). The sample is composed of all municipalities in which mayors are running for reelection that had a first audit during the period 1987-2002.
Columns labeled **A** represent the original results (Bobonis et al., 2016); columns **B** represent the replication results. There are no differences between the results.

Figure 5 shows reelection rates of incumbent mayors as a function of corruption violations by the timing of the audit. Reelection rates seem to decline with more violations, and these effects are larger for timely versus untimely audits. There are no difference between the original and replication results.



(a) Original

(b) Replication

Figure 5: Relationship Between Reported Corruption Levels and Incumbent Mayor's Reelection Rate for Municipalities with Timely and Untimely Audits

Notes: The figure shows the adjusted (by election intercepts) relationship between the mayors who were successfully reelected in the election and the number of corrupt violations per report in the audits for municipalities with a timely and an untimely audit around the period t election.
The original results from Bobonis et al. (2016) are presented next to my replication results (b, d, f) and show no differences.

2.4.2 Audits and Short-Term Electoral Accountability

Table 5 presents linear probability estimates of the reduced-form relationship of audits and electoral outcomes of mayors in both the immediate election (t) and the subsequent election (t+1). Timely audits have no significant impact on average reelection rates in the short-term (column 1), but generate an incumbency advantage in the long-term. Specifically, audits increase long-term reelection rates by 13.3 percentage points (67 percent) for mayors in competitive seats (column 9) and

by 12.8 percentage points (64 percent) for first-term mayors (column 10). Furthermore, the point estimates suggest that timely audits reduce the probability of running for reelection if the mayor receives a negative timely audit (column 5). The replication successfully reproduces these point estimates side-by-side with only a minor rounding error in mean of the dependent variable (column 6). As the authors argue, these dynamics evidence a selection effect: voters select mayors who demonstrate clean audits in period t, providing an incumbency advantage that these mayors use to resume rent-seeking in period $t + 1$.

Table 5: THE EFFECTS OF THE AUDITS ON SHORT-TERM AND LONG-TERM ELECTORAL OUTCOMES

	Incumbent runs for and wins reelection (period t)								Incumbent runs for reelection		Incumbent wins reelection | running	
	OLS (1)		OLS (2)		OLS (3)		OLS (4)		(period t) OLS (5)		(period t) OLS (6)	
	A	B	A	B	A	B	A	B	A	B	A	B
Timely audit	−0.043	−0.043	−0.019	−0.019	−0.002	−0.002	0.031	0.031	−0.011	−0.011	−0.010	−0.010
	(0.040)	(0.040)	(0.056)	(0.056)	(0.052)	(0.052)	(0.053)	(0.053)	(0.065)	(0.065)	(0.056)	(0.056)
Timely audit × num. violations			−0.035	−0.035					−0.068	−0.068	−0.060	−0.060
			(0.029)	(0.029)					(0.042)	(0.042)	(0.028)	(0.028)
Num. of violations			0.000	0.000					−0.013	−0.013	−0.007	−0.007
			(0.014)	(0.014)					(0.014)	(0.014)	(0.013)	(0.013)
Timely audit × incumbent's (t) party has won in previous 3+ elections					−0.076	−0.076						
					(0.080)	(0.080)						
Timely audit × terms in office							−0.070	−0.070				
							(0.034)	(0.034)				
Municipality controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Election year and municipality FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Incumbent mayor (t) FEs	No	No	No	No	No	No	No	No	No	No	No	No
Timely audits F-statistic			1.74	1.74	0.79	0.79	2.68	2.68	1.92	1.92	3.75	3.75
[p-value]			[0.18]	[0.18]	[0.46]	[0.46]	[0.075]	[0.075]	[0.15]	[0.15]	[0.03]	[0.03]
Observations	326	326	326	326	326	326	326	326	326	326	241	241
Mean of dep. variable (controls)	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.76	0.76	0.31	**0.32**

	Incumbent runs for and wins reelection (period $t+1$)											
	OLS (7)		OLS (8)		OLS (9)		OLS (10)		OLS (11)		OLS (12)	
	A	B	A	B	A	B	A	B	A	B	A	B
Timely audit	0.041	0.041	0.080	0.080	0.133	0.133	0.128	0.128	0.050	0.050	0.202	0.202
	(0.055)	(0.055)	(0.066)	(0.066)	(0.072)	(0.072)	(0.072)	(0.072)	(0.068)	(0.068)	(0.074)	(0.074)
Timely audit × num. violations			−0.043	−0.043								
			(0.031)	(0.031)								
Num. of violations			0.007	0.007								
			(0.018)	(0.018)								
Timely audit × incumbent's (t) party has won in previous 3+ elections					−0.179	−0.179			−0.027	−0.027		
					(0.099)	(0.099)			(0.092)	(0.092)		
Timely audit × terms in office							−0.083	−0.083			−0.131	−0.131
							(0.039)	(0.039)			(0.034)	(0.034)
Municipality controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Election year and municipality FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Incumbent mayor (t) FEs	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Timely audits F-statistic			1.07	1.07	2.03	2.03	2.38	2.38	0.27	0.27	7.48	7.48
[p-value]			[0.35]	[0.35]	[0.14]	[0.14]	[0.10]	[0.10]	[0.77]	[0.77]	[0.001]	[0.001]
Observations	326	326	326	326	326	326	326	326	326	326	326	326
Mean of dep. variable (controls)	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20

Notes: Coefficient estimates and standard errors from OLS regressions are presented; disturbance terms are clustered at the municipality level. Controls are the number of municipality government reports, the number of municipal public corporation or consortium reports; indicators for New Progressive Party membership, for incumbent in the opposition party to the state-level executive government, and for incumbent in the opposition party to the governor who appointed comptroller; the vote share for the incumbent in the previous election ($t-1$); and the incumbent's number of terms in office (at time t). The sample is composed of all municipalities that had a first audit during the period 1987–2002. The reported timely audits F-statistic refers to a test of joint significance on the timely audit and its interactions with the relevant explanatory variable (p-value in brackets).
The original results from Bobonis et al. (2016) are presented next to my replication results (b, d, f). Differences between the results are bold.

2.5 Robustness Checks

Table 6 evaluates whether the audit schedule is exogenous by testing for potential political manipulation of the auditing process itself. The authors incorporate controls for deviations from the predetermined audit order, the mayor's political alignment with the governor who appointed the comptroller, and previous electoral win margins. There are no differences between the original and replication results. These results confirm that none of these factors significantly alter the primary short-term or long-term corruption estimates.

Table 6: ROBUSTNESS TESTS: MANIPULATION OF AUDITS

	Number of corrupt violations per report referred to DOJ, in current audit (t)								Number of corrupt violations per report referred to DOJ, in next audit ($t+1$)							
	OLS (1)		OLS (2)		OLS (3)		OLS (4)		OLS (5)		OLS (6)		OLS (7)		OLS (8)	
	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B
Timely audit (period t)	−0.71	−0.71	−0.73	−0.73	−0.48	−0.48	−0.64	−0.64	0.16	0.16	0.13	0.13	−0.09	−0.09	0.34	0.34
	(0.24)	(0.24)	(0.24)	(0.24)	(0.18)	(0.18)	(0.23)	(0.23)	(0.23)	(0.23)	(0.25)	(0.25)	(0.22)	(0.22)	(0.47)	(0.47)
Mean deviation from expected num. of reports (period t)	0.001	0.001							−0.011	−0.011						
	(0.004)	(0.004)							(0.008)	(0.008)						
Mean of abs. value of deviation from expected num. reports (t)			−0.014	−0.014							0.001	0.001				
			(0.013)	(0.013)							(0.017)	(0.017)				
Timely audit × incumbent (period t) from opp. party to governor appointing comptroller					−0.41	−0.41							0.69	0.69		
					(0.37)	(0.37)							(0.33)	(0.33)		
Timely audit × mayor's win margin in previous election, period ($t-1$)							−0.13	−0.13							−1.72	−1.72
							(2.33)	(2.33)							(3.07)	(3.07)
Municipality controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Election year and municipality FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	220	220	220	220	326	326	326	326	198	198	198	198	232	232	232	232
Mean of dep. variable (untimely audits)	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.71	0.71	0.71	0.71	0.75	0.75	0.75	0.75

Notes: Coefficient estimates and standard errors from OLS regressions are presented; disturbance terms are clustered at the municipality level. Controls are the number of municipality government reports, the number of municipal public corporation or consortium reports; indicators for New Progressive Party membership, for incumbent in the opposition party to the state-level executive government, and for incumbent in the opposition party to the governor who appointed comptroller; the vote share for the incumbent in the previous election ($t-1$); and the incumbent's number of terms in office (at time t). The sample is composed of all municipalities that had a first audit during the period 1987–2002.
The original results from Bobonis et al. (2016) are presented next to my replication results (b, d, f). Differences between the results are bolded.

2.5.1 Tests for Alternative Explanations & Additional Robustness Tests

Table 7 conducts a set of robustness checks to rule out alternative mechanisms, such as strategic candidate retirement, the mayor's level of experience, and negative popularity shocks. The authors also introduce incumbent fixed effects (Columns 1 and 5) to isolate the sanctioning effect from potential endogenous sorting of politician types. The authors estimate that timely audits still lead to a decrease of 0.42 corrupt violations per report referred to the DOJ in the short term (column 1). Conversely, in the long term, this specification yields an increase of 0.64 violations per report (column 5).

There is a significant difference in the replication results (column 5) estimating the impact of a timely audit on the number of corrupt violations per report referred to the DOJ in the next audit ($t+1$). This error is due to the replication regression missing a dummy variable for municipalities that received a timely audit in the next election. As such, the replicated results were not distinguishing between the effects of an audit in period t on corruption between those municipalities that have a timely audit in the next period and those who do not. Once I controlled for this variable, I was able to replicate the exact results in the original article.

Table 7: ADDITIONAL ROBUSTNESS TESTS

	Number of corrupt violations per report referred to DOJ, in current audit (t)								Number of corrupt violations per report referred to DOJ, in next audit ($t+1$)									
	OLS (1)		OLS (2)		OLS (3)		OLS (4)		OLS (5)		OLS (6)		OLS (7)		OLS (8)		OLS (9)	
	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B
Timely audit (period t)	−0.42	−0.42	−0.71	−0.71	−0.61	−0.61	−0.81	−0.81	0.64	**0.72**	0.60	0.60	−0.21	−0.21	0.29	0.29	0.84	0.84
	(0.19)	(0.19)	(0.21)	(0.21)	(0.25)	(0.25)	(0.18)	(0.18)	(0.36)	**(0.32)**	(0.48)	(0.48)	(0.57)	(0.57)	(0.23)	(0.23)	(0.36)	(0.36)
Timely audit × terms in office			0.060	0.060														
			(0.12)	(0.12)														
Timely audit, period ($t-1$)					0.04	0.04												
					(0.18)	(0.18)												
Timely audit × shock (period $t+1$) to popularity of party of (period t) incumbent mayor											−0.67	−0.67						
											(0.47)	(0.47)						
Timely audit × incumbent (period $t+1$) runs for reelection													0.40	0.40				
													(0.63)	(0.63)				
Audit period start date							0.02	0.02							0.10	0.10		
							(0.03)	(0.03)							(0.06)	(0.06)		
Audit period end date							0.16	0.16							−0.29	−0.29		
							(0.07)	(0.07)							(0.11)	(0.11)		
Timely audit × incumbent's party has won previous 3+ elections																	−1.43	−1.43
																	(0.53)	(0.53)
Incumbent mayor (t) FEs	Yes	Yes	No	No	No	No	No	No	Yes	Yes	Yes	Yes	No	No	No	No	No	No
Municipality controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Election year and municipality FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	326	326	326	326	326	326	326	326	232	232	232	232	232	232	232	232	232	232
Mean of dep. variable (untimely a.)	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75

Notes: Coefficient estimates and standard errors from OLS regressions are presented; disturbance terms are clustered at the municipality level. Controls are the number of municipality government reports, the number of municipal public corporation or consortium reports; indicators for New Progressive Party membership, for incumbent in the opposition party to the state-level executive government, and for incumbent in the opposition party to the governor who appointed comptroller; the vote share for the incumbent in the previous election ($t-1$); and the incumbent's number of terms in office (at time t). The sample is composed of all municipalities that had a first audit during the period 1987–2002.
The original results from Bobonis et al. (2016) are presented next to my replication results (b, d, f). Differences between the results are bold.

3 Extension 1 - Heterogeneity by Education

3.1 Motivation

Bobonis et al. (2016) document that timely disclosure of municipal audit reports in Puerto Rico reduces corruption in the short run and increases incumbent mayors' subsequent reelection rates. They interpret these findings through a political agency framework in which audits provide information that enables voters both to discipline incumbents (a sanctioning effect) and to select higher-quality politicians (a selection effect). An implicit assumption in this framework is that voters can process and respond to the information contained in audits. However, if voters' ability to interpret and act on audit information varies with local conditions, the effects documented by the authors may be heterogeneous.

This extension examines whether the effects of timely audits on corruption and electoral outcomes vary with municipalities' baseline educational attainment. The hypothesis, grounded in political economy theory, is that more educated voters may be better able to process and understand complex audit findings (Glaeser et al., 2007). Furthermore, as established by Botero, et al. (2013), higher levels of human capital are associated with a greater propensity to demand accountability and report misconduct. Together, these mechanisms suggest that educated voters are better equipped to parse municipal audit reports, attribute responsibility, and update their beliefs regarding incumbent quality (Besley 2006; Ferraz and Finan 2008). If so, the electoral accountability effects documented in the original paper should be stronger in municipalities with higher education levels.

Following the authors' original empirical approach, I use two-way fixed effects models that exploit plausibly exogenous, pre-determined variation in the timing of an audit around mayoral elections between 1987-2005. I extend their baseline specification by interacting the timely audit indicator with municipality-level education measured in 1990. To isolate education from correlated socioeconomic factors, I also estimate specifications that include interactions with baseline municipality median income and poverty rates.

3.2 Methodology

To test whether the effect of timely audit disclosure on corruption outcomes varies with local education, I extend the authors' specification by interacting the timely audit indicator with baseline (1990) education levels (college or more):

$$c_{m,t} = \gamma_t + \alpha_m + \delta_1 \cdot \mathrm{A}_{m,t} + \delta_2 \cdot (A_{m,t} \times Educ_m^{1990}) + \beta X_{m,t} + \varepsilon_{m,t} \tag{5}$$

where $Educ_m^{1990}$ is the share of the adult population with a college education or more, measured in the 1990 US Census, in municipality m. I mean-center this variable so that δ_1 represents the average treatment effect at the sample mean of education. Standard errors are clustered at the municipality level, as in the original study.

Following the authors, I will also estimate analogous specifications for electoral outcomes:

$$e_{m,t} = \gamma_t + \alpha_m + \theta_{E1} A_{m,t} + \theta_{E2}(A_{m,t} \cdot Educ_m^{1990}) + \beta_{E2} X_{m,t} + \epsilon_{m,t} \tag{6}$$

where $e_{m,t}$ indicates whether the incumbent runs for reelection, wins reelection, or wins conditional on running.

To account for the possibility that education proxies for broader socioeconomic conditions, I also estimate the above specifications that include interactions with base-1990 median household income and the poverty rate (mean-centered):

$$c_{m,t} = \gamma_t + \alpha_m + \delta_1 \cdot \mathrm{A}_{m,t} + \delta_2 \cdot (\mathrm{A}_{m,t} \times Educ_m^{1990}) + \delta_3 \cdot (\mathrm{A}_{m,t} \times MedInc_m^{1990}) + \delta_4 \cdot (\mathrm{A}_{m,t} \times Pov_m^{1990}) + \beta X_{m,t} + \varepsilon_{m,t} \tag{7}$$

$$e_{m,t} = \gamma_t + \alpha_m + \theta_{E1} A_{m,t} + \theta_{E2}(A_{m,t} \cdot Educ_m^{1990}) + \theta_{E3} \cdot (\mathrm{A}_{m,t} \times MedInc_m^{1990}) + \theta_{E4} \cdot (\mathrm{A}_{m,t} \times Pov_m^{1990}) + \beta X_{m,t} + \epsilon_{m,t} \tag{8}$$

Equations (5)-(8) are also extended to examine the effect of audits in period t on outcomes in $t+1$.

Because the socioeconomic variables ($Educ_m^{1990}$, $MedInc_m^{1990}$, and Pov_m^{1990}) are measured as pre-determined, cross-sectional baseline characteristics from the 1990 Census, they are absorbed by the municipality fixed effects (α_m). As such, only their interactions with the time-varying timely audit indicator $A_{m,t}$ remain identified in the model. The other variables remain the same as the authors' original specifications. I also report joint F-tests of the socioeconomic interactions to assess whether they moderate the effects of the audits together.

3.3 Results

3.3.1 Effects of the Audits on Corruption

Table 8 reports the effects of audits on short-term corruption outcomes, closely following the structure of the original paper. Columns (1), (3), (5), (7), (9) present baseline estimates with education heterogeneity, while the rest add interactions with base 1990 median income and poverty ratio.

Consistent with the original findings of Bobonis et al. (2016), timely audits significantly decrease corruption in the short-run. For example, column (1) shows that timely audits decrease the number of total violations by 1.384 cases (SE= 0.215), significant at the 1% level. Similar effects are observed for violations involving mayors and vice-mayors (column 2) and those referred to the Department of Justice (column 3). The magnitudes and statistical significance align closely with the original estimates.

Turning to heterogeneity, the interaction between audits and education is generally imprecisely estimated for the number of violations. In column (1), the coefficient on Audit $\times$ Education is 1.914 (SE=4.672), indicating an insignificant moderating effect of education on corruption. Results are similar when examining corruption violations by the the mayor/vice-mayor (column 3). Similarly, in column (3), the interaction is negative but not statistically significant. There is some evidence of heterogeneity, though it is unclear if education is the moderating factor. In column (7), the share of violations involving mayors increases more in higher-education municipalities following audits (coefficient 1.639, SE= 0.690), significant at the 5% level. This suggests that audits may differentially expose high-level corruption in more educated areas. However, when income and poverty interactions are included in column (8), the education interaction becomes smaller and statistically insignificant (0.522, SE = 0.895). The joint F-test for the three interactions is significant, indicating that the socioeconomic environment as a whole moderates this outcome. The individual contributions of any one factor are difficult to examine due to the large degree of collinearity between education, income and poverty.

When income and poverty interactions are added, the main audit effects remain virtually unchanged, reinforcing the robustness of the baseline results. The education interaction becomes less stable and loses statistical significance across most outcomes. Importantly, joint tests of all socioeconomic interactions fail to reject the null for most outcomes, with the exception of column (9), where the joint p-value is 0.008.

Next, I examine the effects of timely audits on the number of corrupt violations in the next electoral cycle (Table 9). The baseline effects of a timely audit on corruption in the next term remain small and statistically insignificant, consistent with the original findings. The education interactions are again insignificant, with the exception of the DOJ share in column10 (3.345, SE=1.698). The joint test for the three interactions is insignificant for all specifications, which suggests an unclear heterogeneous effect of audits on corruption by socioeconomic factors.

Overall, these results suggest that while audits have strong and robust effects on corruption detection in the short-term, and limited effects in the long-term, there is limited and inconsistent evidence that these effects vary systematically with education.

Table 8: Heterogeneous Audit Effects by Education on Short-Term Corruption

	Number of corrupt findings, per report						Share of findings classified as corrupt, findings			
	All		By mayor/vice-mayor		Referred to DOJ		By mayor/vice-mayor		Referred to DOJ	
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)	OLS (7)	OLS (8)	OLS (9)	OLS (10)
Timely audit	-1.384***	-1.378***	-0.638***	-0.641***	-0.536***	-0.531***	-0.075*	-0.072*	-0.120**	-0.126**
	(0.215)	(0.217)	(0.119)	(0.120)	(0.148)	(0.148)	(0.039)	(0.039)	(0.056)	(0.053)
Timely audit × college education+	1.914	-0.580	3.056	3.965	-2.469	-5.110	1.639**	0.522	1.312	2.751
	(4.672)	(6.084)	(2.004)	(2.740)	(3.602)	(4.847)	(0.690)	(0.895)	(1.173)	(1.698)
Timely audit × median income (1,000s)	–	0.107	–	-0.043	–	-0.018	–	0.056	–	-0.157**
		(0.249)		(0.147)		(0.194)		(0.039)		(0.073)
Timely audit × poverty rate	–	0.312	–	-0.176	–	-2.000	–	0.284	–	-1.878
		(4.568)		(2.935)		(3.231)		(0.694)		(1.305)
Municipality controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Election year and mun. FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint p-value	–	0.824	–	0.422	–	0.611	–	0.008	–	0.147
Observations	326	326	326	326	326	326	326	326	326	326
Mean of dep. variable (untimely audits)	2.17	2.17	0.95	0.95	0.99	0.99	0.22	0.22	0.34	0.34

Notes: Coefficient estimates and standard errors from OLS regressions are presented; disturbance terms are clustered at the municipality level. Controls are the number of municipality government reports, the number of municipal public corporation or consortium reports; indicators for New Progressive Party membership, for incumbent in the opposition party to the state-level executive government, and for incumbent in the opposition party to the governor who appointed comptroller; the vote share for the incumbent in the previous election ($t-1$); and the incumbent's number of terms in office (at time t). The sample is composed of all municipalities that had a first audit during the period 1987–2002. Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 9: Heterogeneous Audit Effects by Education on Long-Term Corruption

	Number of corrupt findings, per report						Share of findings classified as corrupt, findings			
	All		By mayor/vice-mayor		Referred to DOJ		By mayor/vice-mayor		Referred to DOJ	
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)	OLS (7)	OLS (8)	OLS (9)	OLS (10)
Timely audit (period t)	-0.147	-0.150	-0.041	-0.040	0.158	0.156	0.058	0.059	0.033	0.033
	(0.292)	(0.289)	(0.142)	(0.143)	(0.191)	(0.188)	(0.058)	(0.054)	(0.065)	(0.062)
Timely audit × college education+	-0.079	-2.468	2.851	2.750	-2.747	-2.047	0.517	0.843	0.808	3.345**
	(4.734)	(8.843)	(3.806)	(4.598)	(3.508)	(5.027)	(1.604)	(1.600)	(1.028)	(1.551)
Timely audit × median income (1,000s)	–	-0.488	–	0.067	–	-0.252	–	0.149**	–	-0.022
		(0.446)		(0.192)		(0.285)		(0.068)		(0.070)
Timely audit × poverty rate	–	-10.881	–	1.232	–	-4.423	–	3.076**	–	1.132
		(6.844)		(3.365)		(4.389)		(1.254)		(1.278)
Municipality controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Election year and mun. FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint p-value	–	0.328	–	0.866	–	0.465	–	0.044	–	0.161
Observations	225	225	225	225	225	225	225	225	225	225
Mean of dep. variable	1.91	1.91	0.83	0.83	0.75	0.75	0.21	0.21	0.36	0.36

Notes: Results for corruption outcomes in the subsequent election period ($t+1$) . Centered socioeconomic variables (1990) are used for interactions. Controls include reporting frequency, political party indicators, and incumbent experience. Standard errors clustered at the municipality level. * p < 0.10, ** p < 0.05, *** p < 0.01.

3.3.2 Electoral Outcomes

Table 10 examines whether the electoral consequences of audits vary with education. Consistent with the original paper, the baseline effect of audits on electoral outcomes is small and statistically insignificant. For example, in column (1), the effect of audits on incumbent re-election is 0.046 (SE = 0.055), indicating no meaningful electoral penalty. In contrast, the interaction between audits and education is positive and this masks some heterogeneity. The point estimate suggests that the probability of a successful reelection is 1.99 percentage points higher for a marginal increase in the percent of college educated (or higher) than the average level (column 1). Incumbent mayors are also more likely to run for reelection (column 3) the more educated on average their municipality is. However, among those running, their reelection probability is (insignificantly) higher in the current (column 5) election. Their reelection probability is (insignificantly) weaker in the next period election,

however (column 7). These results suggest that in more educated municipalities, audits may increase political engagement or responsiveness, potentially encouraging incumbents to run again or improving their electoral prospects. However, these effects are not consistently observed across all outcomes.

When additional socioeconomic interactions are included (columns 2, 4, 6, 8), the estimated education effects become larger in magnitude and more precisely estimated in some cases (e.g., column 2: 5.242, SE=1.684), but the instability across specifications suggests caution in interpretation. The joint F-test for the inclusion of the three socioeconomic interactions is significant in some cases (column 2, 8), suggesting that the socioeconomic environment moderates the probability of reelection due to the audits. However, the collinearity among socioeconomic measures makes it difficult to isolate the role of education specifically.

Overall, these findings provides some support for the hypothesis that education strengthens electoral accountability. While some coefficients are statistically significant, the results are not consistent across outcomes or specifications.

Table 10: The Effects of the Audits on Short-Term and Long-Term Electoral Outcomes

	Incumbent runs for and wins reelection (period t)		Incumbent runs for reelection (period t)		Incumbent wins reelection | running (period t)		Incumbent runs for and wins reelection (period $t+1$)	
	OLS (1)	OLS (2)	OLS (3)	OLS (4)	OLS (5)	OLS (6)	OLS (7)	OLS (8)
Timely audit	-0.046	-0.056	-0.078	-0.084	-0.052	-0.044	0.003	0.004
	(0.055)	(0.051)	(0.062)	(0.060)	(0.066)	(0.064)	(0.056)	(0.055)
Timely audit × college education+	1.990*	5.242***	2.831**	4.198**	0.130	2.768	-2.369	-1.651
	(1.127)	(1.684)	(1.180)	(1.919)	(1.198)	(2.095)	(1.545)	(1.739)
Timely audit × median income (1,000s)	–	-0.191**	–	-0.175**	–	-0.067	–	0.131**
		(0.075)		(0.081)		(0.082)		(0.062)
Timely audit × poverty rate	–	-1.313	–	-2.238	–	0.228	–	2.794**
		(1.174)		(1.464)		(1.593)		(1.117)
Municipality controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Election year and mun. FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Joint SES p-value	–	0.013	–	0.073	–	0.496	–	0.042
Observations	326	326	326	326	238	238	326	326
Mean of dep. variable (controls)	0.25	0.25	0.76	0.76	0.32	0.32	0.20	0.20

Notes: Coefficient estimates and standard errors from OLS regressions are presented; disturbance terms are clustered at the municipality level. Controls are the number of municipality government reports, the number of municipal public corporation or consortium reports; indicators for New Progressive Party membership, for incumbent in the opposition party to the state-level executive government, and for incumbent in the opposition party to the governor who appointed comptroller; the vote share for the incumbent in the previous election ($t1$); and the incumbent's number of terms in office (at time t). All base socioeconomic variables are mean-centered. The sample is composed of all municipalities that had a first audit during the period 1987–2002. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.4 Discussion

The results of this extension provide limited evidence that education moderates responses to audits. While some interaction coefficients are statistically significant, the pattern is not consistent across specifications, and many estimates are imprecise. Overall, the evidence suggests that any heterogeneity by education is modest and sensitive to alternative specifications. These findings should therefore be interpreted cautiously.

Several limitations should be noted. First, the TWFE specification may mask heterogeneity across treatment cohorts and time, particularly in settings with staggered and non-absorbing treatment timing (de Chaisemartin and D'Haultfoeuille, 2020). The next extension will address these concerns with respect to the authors' main findings. Second, education is measured at baseline and does not vary over time, so the interaction with the audit variable relies on cross-sectional variation that may correlate with unobserved characteristics. Third, including multiple correlated socioeconomic interactions introduces

multicollinearity. Finally, the electoral results are noisier and may lack statistical power, especially for subsamples like incumbents who run for re-election.

4 Extension 2 - Modern Difference-in-Differences

The empirical strategy in Bobonis et al. (2016) uses two-way fixed effects (TWFE) models. This extension re-examines the dynamic effects of audit timing using a modern difference-in-differences estimator that accommodates non-absorbing treatments. In this setting, municipalities may experience both timely and untimely audits across different election cycles, meaning treatment status can change over time. This violates the standard staggered adoption framework assumed by traditional two-way fixed effects estimators (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfoeuille, 2020). In settings with staggered treatment adoption, TWFE implicitly uses already-treated units as comparisons for later-treated units. If treatment effects evolve over time, these comparisons can receive negative weights and bias the estimated average treatment effect.

To address this, I implement a dynamic difference-in-differences estimator ("modern DiD") that remains valid when treatment status is staggered and non-absorbing (de Chaisemartin and D'Haultfoeuille, 2020, 2025) . This approach allows for more flexible dynamic treatment effects and avoids biases associated with negative weighting in standard estimators. The authors' theoretical model predicts that treatment effects may be dynamic: politicians reduce corruption before an election but may increase it afterward if they are not scheduled for another timely audit. Under these conditions, the TWFE estimator may be biased.

4.1 Methodology

To test the robustness of the original results and account for potential biases in the TWFE estimator when treatment timing varies and effects are dynamic, I re-estimate the effects of timely audits using the dynamic difference-in-differences estimator developed by de Chaisemartin and D'Haultfoeuille (2020, 2025). This estimator is robust to heterogeneous and dynamic treatment effects and avoids the negative weighting issues that can arise in TWFE when already-treated units serve as controls for later-treated units.

The estimator constructs treatment effects by comparing municipalities that newly receive a timely audit in a given election cycle with municipalities that have not yet been treated at that time. For a given cohort g (municipalities first treated in period t), the treatment effect is:

$$\hat{\delta}(g,t) = (Y_{g,t} - Y_{g,t-1}) - \frac{1}{|N_{g,t}|} \sum_{i \in C_{g,t}} (Y_{i,t} - Y_{i,t-1}) \tag{9}$$

where $C_{g,t}$ is the set of municipalities that have not been treated yet by period t, and Y is a particular outcome (e.g., corruption). The estimator then aggregates these cohort-specific differences-in-differences using weights proportional to cohort size and treatment variance, yielding an average treatment effect for the first period of treatment. Dynamic effects (e.g., effects one period after treatment) are obtained by extending the comparison window accordingly.

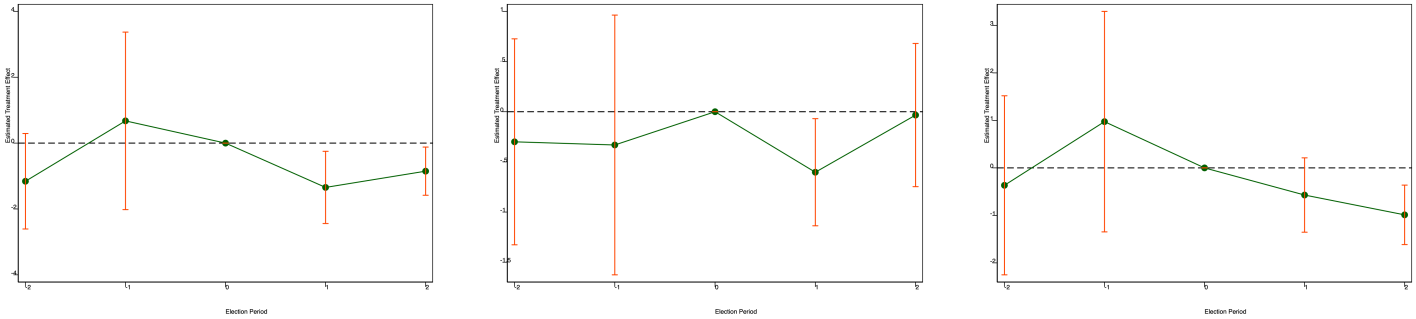
The estimator allows treatment to switch on and off across periods. This is a key feature for this setting because a municipality that receives a timely audit in period $t$may receive either a timely or untimely audit in period $t+1$. Following Bobonis et al. (2016), I include the same set of political controls: the number of audit reports, indicators for New Progressive Party membership, indicators for whether the incumbent is in opposition to the state-level executive government or to the governor who appointed the comptroller, the incumbent's vote share in the previous election, and the incumbent's number of terms in office. The estimator handles these time-varying covariates by residualizing the outcomes with respect to them before computing the treatment effects.

The validity of the estimator relies on two assumptions: no anticipation (treatment effects begin only after treatment) and parallel trends conditional on covariates. I test these using placebo estimates in the period before treatment (leads). If significant effects appear before treatment, that would indicate violations. Bobonis et al. (2016) provide graphical evidence (Figure 4) that corruption levels were similar across timely and untimely audit municipalities in earlier periods; the placebo tests in the modern DiD framework offer a formal check of this assumption.

4.2 Results

In this section, I examine whether the results Bobonis et al. (2016) found to answer the following question are robust to the use of a more modern estimation: i) What are the short- and long-run effects of the audit program on corruption? and, ii) How does the program impact electoral outcomes?

4.2.1 Dynamic Event Study Graphs



(a) All findings (b) Findings by mayor or vice-mayor (c) Findings referred to PR DOJ

Figure 6: Number of Findings Across Time, by Timeliness of Audit in Election t

Notes: This figure plots the estimated dynamic treatment effects of a timely audit relative to the year of the audit ($t = 0$). 95% Confidence Intervals are displayed in red.

Figure 6 displays the dynamic treatment effects for the primary corruption outcomes across the entire period of study ($1987 - 2005$). The pre-treatment placebos up to two periods ($t = -2, -1$) before a timely audit ($t = 0$) are consistently near zero and statistically insignificant, which suggests not-yet-treated and untreated municipalities were trending similarly before a timely audit ($t = 0$). Note, however, the confidence intervals for the placebo estimates are wide; this is a consequence of the dynamic estimator's strict restriction of treatment and control groups, reducing statistical power. As such, while we fail to reject the null hypothesis of parallel trends, the wide confidence intervals suggest the test may be underpowered to detect subtle, pre-existing deviations in municipal corruption trajectories. Despite this limitation, the statistically significant treatment effects in the short- ($t = 1$) and long-run ($t = 2$) provide evidence that the timely audit impacted corruption outcomes.

The treatment effect trajectories in Figure 6 align closely with the results in Bobonis et al. (2016). Panel A illustrates the short-run ($t = 1$) sanctioning effect on all corruption violations; long-run ($t = 2$), these effects appear less pronounced. Panel B shows that, while mayors/vice-mayors reduce their corrupt activities in the immediate short-run following an audit ($t = 1$), this effect fades entirely by the subsequent election cycle ($t = 2$). However, Panel C challenges the original authors' results. Severe corruption violations (referred to the DOJ) are not significantly impacted in the short-run ($t = 1$), but decline significantly in the long-run ($t = 2$).

The following sections discuss the results of applying the dynamic DiD estimator to the authors' main findings.

4.2.2 Short-Term Effects of the Audits on Corruption

Table 11, panel A presents estimates of the short-term effects of timely audits on rent-seeking levels using the dynamic DiD estimator. Column 1 shows that a timely audit reduces the total number of corrupt violations per report by -1.345 (SE = 0.559), significant at the 5% level. This estimate is slightly smaller in magnitude than the TWFE estimate of -1.43 (SE = 0.22) (Table 3, Panel A, column 1), but the confidence intervals overlap substantially. The 95% confidence interval for the modern DiD estimate contains the TWFE point estimate.

For mayor/vice-mayor violations (column 2), the estimate is -0.603 (SE = 0.272), significant at the 5% level, compared to the TWFE's estimate of -0.63 (SE = 0.14). For DOJ referrals (column 3), the estimate is -0.571 (SE = 0.399), which is not statistically significant, whereas Bobonis et al. (2016)'s estimate of -0.65 (SE = 0.16) is significant. The loss of precision for DOJ referrals likely reflects the smaller number of "switchers" (municipalities that experience a change in audit status) used to identify the effect (55 municipalities in this sample versus 326 municipality-election observations in the original).

For the share of corruption violations (columns 4-5), neither estimate is statistically significant, and both are smaller in magnitude than the TWFE estimates. The share of findings attributed to the mayor/vice-mayor shows a positive and insignificant effect (0.018, SE= 0.089) compared to the TWFE estimate of -0.082 (SE = 0.040); the share referred to DOJ shows -0.172 (SE= 0.109) compared to the TWFE one (-0.160; SE= 0.055).

The placebo tests show no statistically significant effects in the period before treatment for any outcome, providing support for the parallel trends assumption. However, these tests are based on smaller samples (59 observations, 31 switchers)

and are imprecisely estimated.

Table 11: Short-Term and Long-Term Effects of Audits on Corruption (Modern DiD)

	Number of corrupt violations, per report			Share of findings classified as corrupt violations	
	All (1)	By Mayor/ Vice-Mayor (2)	Referred to DOJ (3)	By Mayor/ Vice-Mayor (4)	Referred to DOJ (5)
Panel A: Short-Term Outcomes (in period t)					
Timely Audit (t)	-1.345^{**}	-0.603^{**}	-0.571	0.018	-0.172
	(0.559)	(0.272)	(0.399)	(0.089)	(0.109)
Observations	128	128	128	128	128
Switchers	55	55	55	55	55
Panel B: Long-Term Outcomes (in period $t+1$)					
Timely Audit (t)	-0.852^{**}	-0.033	-0.989^{***}	0.156	-0.338^{***}
	(0.373)	(0.364)	(0.319)	(0.147)	(0.129)
Observations	85	85	85	85	85
Switchers	42	42	42	42	42
Joint Null Effect p-value	0.0071^{***}	0.065^{*}	0.0073^{***}	0.36	0.0076^{***}
Municipality Controls	Yes	Yes	Yes	Yes	Yes
Election year and municipality FEs	Yes	Yes	Yes	Yes	Yes
Placebo Effect ($t-1$)	0.675	-0.331	0.974	-0.014	0.136
	(1.375)	(0.660)	(1.184)	(0.121)	(0.302)
Observations	59	59	59	59	59
Switchers	31	31	31	31	31

Notes: Coefficient estimates and standard errors from OLS regressions are presented (de Chaisemartin and D'Haultfœuille, 2020, 2025), which is robust to heterogeneous and dynamic treatment effects. Standard errors (in parentheses) are clustered at the municipality level. Controls are the number of municipality government reports, the number of municipal public corporation or consortium reports; indicators for New Progressive Party membership, for incumbent in the opposition party to the state-level executive government, and for incumbent in the opposition party to the governor who appointed comptroller; the vote share for the incumbent in the previous election ($t-1$); and the incumbent's number of terms in office (at time t). The sample is composed of all municipalities that had a first audit during the period 1987–2002.
Panel A and B report coefficients of the effect of an audit (at time t) on immediate (t) and subsequent ($t+1$) outcomes. Identification is driven by the municipalities (switchers) that experience a transition in audit status during the sample period. Panel C reports the first-order placebo to test the parallel trends assumption; the Joint Null Effect p-value reports a Wald test for the hypothesis that all treatment effects are jointly zero. $*p < 0.10, **p < 0.05, ***p < 0.01$.

4.2.3 Long-Term Effects of the Audits on Corruption

Next, I estimate the effect of timely audits on the number and share of corrupt violations in the subsequent term ($t+1$) to examine the audit's long-term effects (Table 11, panel B) and compare them to the original results (Table 4). Using TWFE, the authors find that these effects are small and statistically insignificant: -0.10 for total violations, -0.07 for mayor-level violations, and 0.16 for DOJ referrals (Table 4, Panel A).

The modern DiD estimates suggest larger long-term reductions in corruption than the TWFE estimates. For all violations (Table 11, column 1), the estimated effect is -0.852 (SE= 0.373), significant at the 5% level. For DOJ referrals (column 3), the estimate is -0.989 (SE= 0.319), significant at the 1% level. For mayor-level violations (column 2), the estimate is close to zero and insignificant (-0.033, SE= 0.364). The joint test that all five long-term effects are zero is strongly rejected (p = 0.007). These results suggest that the long-term effects of timely audits on rent-seeking levels may be more pronounced than the TWFE estimates indicate, particularly for severe violations referred to the DOJ. These results would suggest a more persistent effect than Bobonis et al. (2016) found, but they should be interpreted with caution due to the smaller sample of switchers.

Several caveats are in order to interpret these results. First, the sample for long-term effects is smaller (85 observations, 42 switchers) than the original TWFE sample (232 observations); this under-powers the modern DiD estimation. Second, the modern DiD estimator identifies effects using only municipalities that switch treatment status, whereas the TWFE estimator uses all observations. If the municipalities that switch are systematically different from those that do not, the estimates may

not be comparable. Third, the long-term effects are identified off a different comparison group than the short-term effects - i.e., municipalities that experience a timely audit in period t are compared to those that have never received a timely audit by period $t + 1$, which may be a selected sample.

Given these considerations, the most appropriate interpretation is that the original findings are broadly robust to the use of a more flexible estimator, though some discrepancies suggest that the original estimates may mask heterogeneity or dynamics.

4.2.4 Short-Term Effects of the Audit on Electoral Outcomes

Table 12, panel A presents modern DiD estimates of the reduced-form relationships between timely audits and electoral outcomes in the current electoral term. The authors emphasize that these estimates are not causal effects of the timely audit content on electoral outcomes due to the endogeneity of the audit outcomes. This limitation also applies to these results.

The unconditional effects on the probability of an incumbent running (-0.256, SE= 0.163) or winning (-0.060, SE= 0.080) reelection remain negative but statistically insignificant (column 1-2). The result is similar when examining the conditional effect "Win | Run" (column 3). This specification reduces the valid cohort of "switchers" (i.e., municipalities that transitioned from untimely to a timely audit) from 55 to 32, a loss of over 40% of the treated group. As such, this result may capture a combination of voter sanctioning of corrupt mayors as well as an endogenous, strategic exit of the most corruptible mayors, which artificially restricts the sample to a more positively selected subgroup.

The placebo tests show no significant pre-trends, though they are based on very small samples (17-31 switchers).

4.2.5 Long-Term Effects of the Audit on Electoral Outcomes

Table 12, panel B reports effects on electoral outcomes in the subsequent election term (period $t+1$). A timely audit is linked to a -0.096 (SE= 0.085) effect on the probability of incumbent winning re-election (column 1), compared to the TWFE's positive but insignificant 0.041 (SE= 0.055) estimate (Table 5, column 7).

For the probability that the incumbent runs (column 2), the estimate is -0.354 (SE= 0.117), significant at the 1% level, compared to the TWFE's insignificant 0.080 (SE= 0.066) (Table 5, column 8). This large negative effect suggests that timely audits may discourage incumbents from seeking reelection in the subsequent term, which is not a result found in the original study.

For the probability of winning conditional on running (column 3), the estimate is essentially zero (-0.000, SE= 0.131), compared to TWFE's 0.133 (SE= 0.072) estimate (Table 5, column 9).

Table 12: The Effects of the Audits on Short-Term and Long-Term Electoral Outcomes (Modern DiD)

	(1) Incumbent runs for and wins reelection (t)	(2) Incumbent runs for reelection (t)	(3) Incumbent wins reelection | running (t)
Panel A: Short-Term Outcomes (in period t)			
Timely Audit (t)	-0.060	-0.256	-0.100
	(0.080)	(0.163)	(0.079)
Observations	128	128	72
Switchers	55	55	32
Panel B: Long-Term Outcomes (in period $t+1$)			
Timely Audit (t)	-0.096	-0.354***	-0.000
	(0.085)	(0.117)	(0.131)
Observations	85	85	42
Switchers	42	42	19
Joint Null Effect p-value	0.051*	0.011**	0.42
Municipality Controls	Yes	Yes	Yes
Election year and municipality FEs	Yes	Yes	Yes
Placebo Effect ($t-1$)	0.026	-0.034	0.136
	(0.121)	(0.256)	(0.104)
Observations	59	59	17
Switchers	31	31	11

Notes: Coefficient estimates and standard errors from OLS regressions are presented (de Chaisemartin and D'Haultfœuille, 2020, 2025), which is robust to heterogeneous and dynamic treatment effects. Standard errors (in parentheses) are clustered at the municipality level. Controls are the number of municipality government reports, the number of municipal public corporation or consortium reports; indicators for New Progressive Party membership, for incumbent in the opposition party to the state-level executive government, and for incumbent in the opposition party to the governor who appointed comptroller; the vote share for the incumbent in the previous election ($t-1$); and the incumbent's number of terms in office (at time t). The sample is composed of all municipalities that had a first audit during the period 1987–2002.
Panel A and B report coefficients of the effect of an audit (at time t) on immediate (t) and subsequent ($t+1$) electoral outcomes. Identification is driven by the municipalities (switchers) that experience a transition in audit status during the sample period. Panel C reports the first-order placebo to test the parallel trends assumption. The Joint Null Effect p-value reports a test for the hypothesis that all treatment effects in period t and $t+1$ are jointly zero.
$*p < 0.10, **p < 0.05, ***p < 0.01$.

4.3 Discussion

The modern staggered difference-in-differences estimates provide a rigorous test of Bobonis et al. (2016)'s findings, providing results that both confirm and challenge the original TWFE conclusions.

First, the modern DiD estimates broadly confirm the core finding that timely audits significantly reduce corruption in the short run. Point estimates for total violations and mayor-level violations are similar in magnitude to the original TWFE results. The estimates diverge, however, regarding long-term corruption. While the original authors find that corruption rebounds in the subsequent term, the dynamic estimator suggests a more pronounced, sustained reduction, particularly for severe violations referred to the DOJ.

There are several limitations in the interpretation of these results. The dynamic DiD estimator identifies effects only from municipalities that switch treatment status, significantly reducing sample size and power. For instance, the sample declines from 55 to 32 switchers when conditioning on incumbents who run for reelection, introducing selection bias because the decision to run is endogenous to the audit outcome. Moreover, replication of the continuous audit timing manipulation and tests of alternative theoretical mechanisms (Tables 6-7) are omitted here, since the panel data is too sparse for the dynamic DiD estimator. As such, these results cannot be adequately checked in this extension.

5 Conclusion

This paper successfully replicates the core findings of Bobonis et al. (2016). The results show that timely audits reduce short-run corruption following timely audits, but do not lead to sustained effects in the subsequent term. Furthermore, timely audits increase incumbent mayors' electoral accountability. The only notable discrepancy was a coding omission in one robustness table, which did not affect the main conclusions.

I extend the original paper in two significant ways. The first extension examines whether the effects of audits vary with municipal education levels. The results yield no consistent evidence of educational heterogeneity regarding corruption outcomes, supporting the interpretation that these initial effects operate primarily through politicians' anticipatory responses to voter sanctions rather than varying voter sophistication. Electoral outcomes show some suggestive heterogeneity: timely audits are associated with higher probabilities of incumbents running for reelection and, in some specifications, higher probabilities of success in more educated municipalities. However, these patterns are not robust to the inclusion of income and poverty interactions, and the high collinearity among socioeconomic measures makes it difficult to isolate the role of education. Overall, the evidence for educational heterogeneity is limited and inconclusive.

The second extension re-estimates the main results using a dynamic difference-in-differences estimator robust to heterogeneous and dynamic treatment effects, which are plausible in this setting. The short-run corruption effect is robust, with point estimates similar to the original TWFE results. In contrast, the long-run results differ: the dynamic estimator suggests larger, statistically significant reductions in total violations and DOJ referrals in the subsequent term, whereas the original TWFE found no sustained effect. The electoral results are less robust, with the dynamic estimator producing different point estimates and highlighting severe sample attrition when conditioning on the decision to run. These differences underscore that conclusions about long-run dynamics and electoral accountability may be sensitive to the choice of estimator, especially given the small number of municipalities that change treatment status.

Together, the extensions confirm the original paper's central insight that timely audits discipline politicians in the short run, but they also reveal that the longer-term patterns are more complex. These results reinforce the importance of both institutional context and methodological rigor in evaluating political accountability.

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